Casa Ericeira - HVAC + Water Systems Spec Factsheet

Project context



- New build, 180 m^2 across 2 floors
- Ground floor (~90 m^2): open plan kitchen + living + hallway, flat 2.5m ceiling. **Climatized area ~64 m^2**
  - Kitchen and Meals: ~25 m^2
  - Living: ~30 m^2
  - Entrance and Hallway (open to living): ~10 m^2
- First floor (~90 m^2): 2 bedrooms + 2 bathrooms + walk-in closet, vaulted ceiling (2.5m flat + 1.5m sloped V with apparent beams). **Climatized area ~34 m^2**
  - Bedroom 1: ~16 m^2
  - Bedroom 2: ~15 m^2
  - Circulation: ~3 m^2
- Underground concrete garage (technical zone for all equipment)
- Wood frame structure above garage
- 2 occupants regularly, 4-5 occasionally. Location: google maps.
-

HVAC priorities

- 4-zone individual room control (Kitchen, Living, Bedroom 1, Bedroom 2)
- Clean aesthetics: ceiling diffusers only, no wall-mounted units
- Preserve vaulted ceilings in bedrooms (no false ceiling under the V)
- Single technical zone in garage for all water equipment

1. HVAC Configuration

Heat pumps (ducted, 4-zone control)

Suggested: Mitsubishi Electric
- Ground floor unit: PEAD-M71JA (7.1 kW), suspended from garage ceiling
- First floor unit: PEAD-M50JA (5.0 kW), installed in attic above circulation hallway or walk-in closet area (preferred: circulation hallway for cleaner maintenance access without disturbing closet contents - to be confirmed with architect based on attic clearance)
- 2 outdoor units mono-split SUZ-M71VA + SUZ-M50VA, grouped at north facade
- 2x PAC-ZC40H-E zone controllers (4 zones total, 240V dampers)

Centralized HRV (Heat Recovery Ventilation)

Enthalpy recovery non-negotiable for humidity management in marine climate.

Suggested: Mitsubishi Lossnay VL-250 - 250 m^3/h, 84% heat recovery, native integration with Mitsubishi PEAD - Single contractor and MELCloud app for AC + HRV

Diffuser schedule

Ground floor (validated placement on floor plan):

Code	Type	Location	Function
L1	Linear slot vertical	Kitchen, above island, west wall	Supply Kitchen zone
S1	Square ceiling 300x300mm	Kitchen, near south bay window (dining area)	Supply Kitchen zone
R1	Return grille	Kitchen, near hallway entrance	Return Kitchen zone
L2	Linear slot vertical	Living, between entrance and living	Supply Living zone
S2	Square ceiling 300x300mm	Living, near south wall	Supply Living zone
R2	Return grille	Living, NE corner near armchairs	Return Living zone

Total RDC: 4 supply + 2 return.

First floor (per bedroom, vaulted ceiling):

Element	Type	Location
Supply	Linear slot 600x100mm	Flat ceiling section at 2.5m, near window
Return	Grille 300x150mm	Wall, low level (30-50cm from floor), opposite supply

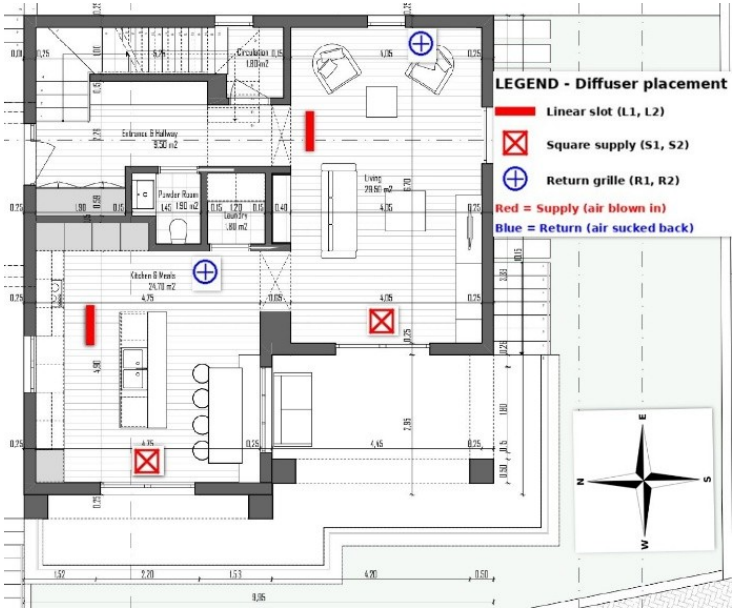
Low-level wall returns critical to force vertical air sweep under the V profile and avoid short-circuiting.

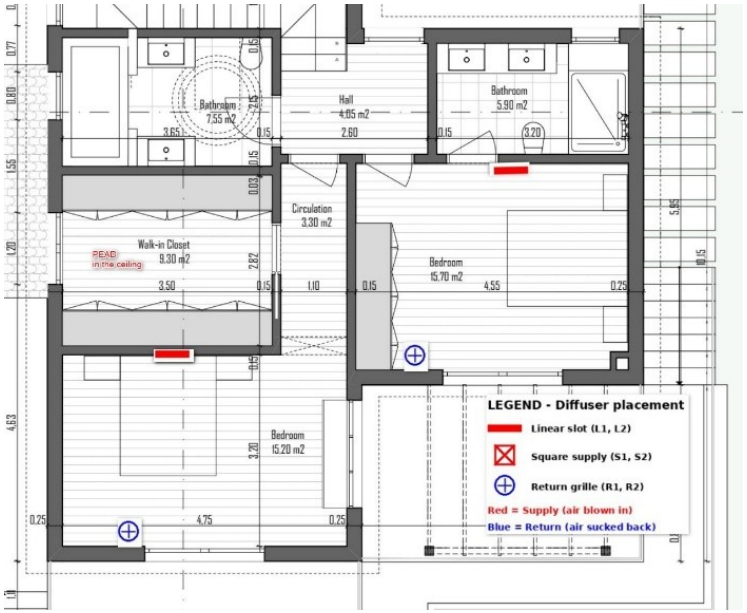
Walk-in closet and circulation hallway (flexibility provisions):

Zone	HRV	AC	Purpose
Walk-in closet (~9 m^2)	Supply + extraction	**Duct provision only** (no diffuser installed)	Prevent textile humidity/odors; allow future conversion to bedroom/office without rework
Circulation hallway (~3 m^2)	Transfer grille or low extraction	Air overflow from adjacent zones	Maintain air circulation between bedrooms and bathrooms

Walk-in closet may eventually be converted into a child's bedroom or home office. Reserving an AC duct path now avoids a major retrofit later if attic access becomes constrained.

No ceiling fans (incompatible with V ceiling and apparent beams).
No floor diffusers (cold air stagnation in cooling mode unsuitable for marine climate).





Architectural provisions for HVAC

- Slab penetration sleeves between garage and ground floor: 8-10 sleeves Ø150-200mm, cast into formwork BEFORE concrete pour (irreversible)
- Attic clearance above circulation hallway / walk-in closet: minimum 300mm above PEAD-M50JA unit
- False ceiling at RDC: required throughout to route AC + HRV ducts
- **False ceiling in walk-in closet AND circulation hallway**: required for AC + HRV + plumbing routing (boucle recirculation manifold)
- Outdoor unit positions on north facade with refrigerant pipe routing to each indoor unit
- 230V supply provisions: 2x indoor units + 1x HRV unit + 2x outdoor units

2. Water Systems Configuration

Domestic hot water (PAC ECS)

Suggested: Ariston Nuos Plus S2 WI-FI 250L (Class A+) - COP 3.35, 333L water at 40°C in single extraction - Titanium tank, WiFi built-in, mode SILENT 52 dB(A) - Floor-standing in garage

Hot water recirculation (demand-controlled, push-button) - OPTIONAL

Suggested: Grundfos Comfort PM AutoAdapt (15-14 BX PM) - ECM motor, A class, AutoAdapt mode learns usage - Brass body for hot water network - Permanent magnet motor for low consumption

Configuration:
- 2 push-buttons (Bathroom Master + Bathroom Bedroom 1), IP44 with LED, near bathroom entrance
- Loop "terminating short of fixtures" design: loop ends at central manifold in plenum above walk-in closet - Short branches (~2m) from manifold to each upper bathroom
- Insulated multilayer pipe Ø10-16mm

Sediment filter (pre-filtration, Phase 1)

Suggested: Atlas Filtri or Honeywell - 5-100 micron washable filter, brass housing, pressure gauge - Installed on main water inlet, protects PAC ECS and entire plumbing

Water softener (Phase 2 - provisions only now)

Phase 1 provisions (now): - T-junction + bypass valve on main water inlet - Isolation valves (in and out) - Floor drain within ~1m for regeneration waste water - 220V outlet near future install location - Reserved space ~50x50x120 cm near water inlet

Phase 2 (later, if hardness test justifies install):

Suggested: Bosch Aqua 4000 S / 8000i S - Wide Bosch SAV network across Portugal - HomeCom Easy WiFi app with diagnostics and salt alerts - Adjustable treatment level

Architectural provisions for water

- Reserved technical zone in garage for: PAC ECS (1m^3 free air around unit), softener provisions, sediment filter, recirculation pump
- Manifold point in plenum above walk-in closet (recirculation loop terminus, equidistant from both upper bathrooms)
- 220V outlet near future softener location
- Floor drain near future softener location
- Push-button cable routing from each upper bathroom to recirculation pump at garage

3. Combined Layout Summary

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GARAGE (technical zone)
   ├── PAC RDC (PEAD-M71JA suspended from slab ceiling)
   │    └── 5-6 slab penetrations → RDC false ceiling
   ├── PAC ECS (250L, floor-standing)
   ├── Grundfos recirculation pump
   ├── Sediment filter on main inlet
   ├── Provisions for future water softener
   └── 2 outdoor AC units at north facade

GROUND FLOOR (RDC)
   ├── Kitchen zone: L1 + S1 supply, R1 return
   ├── Living zone: L2 + S2 supply, R2 return
   └── Powder room: direct hot water supply

ATTIC (above circulation hallway or walk-in closet)
   ├── PAC étage (PEAD-M50JA) - preferred: above circulation hallway
   ├── HRV unit (Lossnay VL-250 or equivalent)
   └── Recirculation manifold (hot water loop terminus)

FIRST FLOOR (étage)
   ├── Bedroom 1: linear slot supply (ceiling 2.5m), low wall return
   ├── Bedroom 2: linear slot supply (ceiling 2.5m), low wall return
   ├── Walk-in closet: HRV supply + extraction, AC duct provision (reserved for future use)
   ├── Circulation hallway: HRV transfer, false ceiling for routing
   ├── Bathroom Master: push-button + short hot water branch
   ├── Bathroom Bedroom 1: push-button + short hot water branch
   └── HRV extracts in bathrooms, supplies in bedrooms + walk-in
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4. Questions for Engineers and Installers

Architect

1. Slab penetration sleeves between garage and RDC: 8-10 sleeves Ø150-200mm cast into formwork BEFORE concrete pour - positions to define with HVAC engineer
2. **PAC 1st Floor**: preferred position above the circulation hallway (cleaner maintenance access without disturbing closet contents); confirm attic clearance is suitable in both options (hallway and walk-in)
3. **False ceiling required in walk-in closet AND circulation hallway** for HVAC + HRV + plumbing routing (recirculation manifold)
4. **Walk-in closet ventilation and conversion-ready design**: confirm HRV supply + extraction installed, plus AC duct provision (reserved but not installed) for future room conversion (kid bedroom, office, guest room)
5. Vaulted bedroom ceiling profile: confirm flat section at 2.5m is wide enough for linear slot diffuser boxes
6. False ceiling required at RDC for AC + HRV ducts
7. Wall cavities for HRV ducts and low-level return ducts in bedrooms (wood frame structure)
8. Outdoor unit positions on north facade: visual integration and refrigerant pipe routing
9. HRV exterior wall penetrations: fresh air intake + exhaust positions on facade
10. Garage technical zone layout: reserve space for PAC ECS (~600x600x1500mm), sediment filter, softener provisions, recirculation pump
11. Acoustic isolation in attic above bedrooms (silent blocks for PEAD-M50JA, anti-vibration mounts)
12. Cable routing for push-buttons from upper bathrooms to garage (recirculation pump)

HVAC engineer

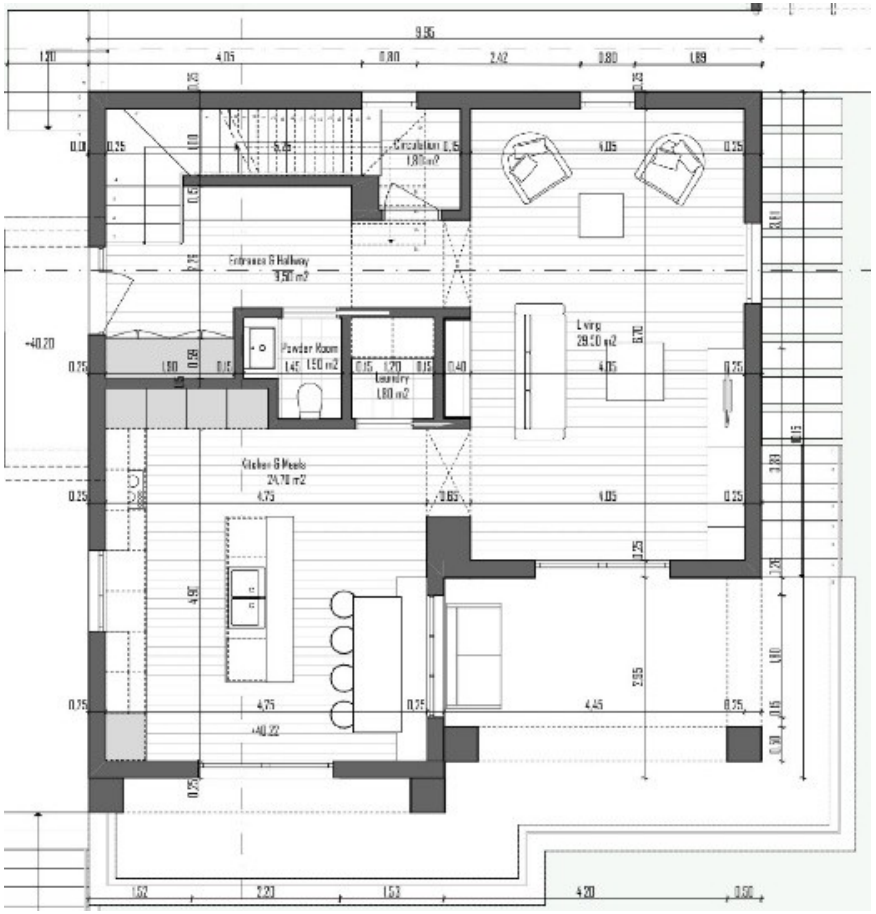
1. Confirm thermal load calculation: validates PEAD-M71JA (RDC) + PEAD-M50JA (étage) sizing
2. Beam spacing in vaulted bedrooms: coordinate diffuser positions
3. False ceiling void depth needed at RDC for AC + HRV ducts in parallel
4. Recommendation: ceiling diffusers confirmed over floor grilles for marine climate
5. Pressure static requirements based on duct length from garage to RDC and from attic to bedrooms
6. HRV location: in attic or in garage? Optimal for duct routing and noise
7. Confirm refrigerant pipe routing from outdoor units (north facade) to both indoor units
8. Bedroom returns at low level (30-50cm from floor): confirm wall cavity routing in wood frame
9. Diffuser brand recommendation: linear slot models compatible with the AC airflow

Plumber

1. PAC ECS placement: confirm 1m^3 free air clearance and condensate drain location
2. Recirculation loop terminus: confirm manifold point in plenum above walk-in closet
3. Loop length measurement: PAC ECS → manifold → return path
4. Push-button positioning and IP44 compliance near bathroom entrances
5. Push-button cable routing through wood frame walls

6. Future softener provisions: T-junction size (1” or 1.5”), bypass valve, drain, 220V outlet
7. Hose bibs (garden taps): keep upstream of softener provisions (raw water)
8. Anti-legionella cycle on PAC ECS: confirm enabled
9. PV-readiness on PAC ECS (Smart Grid Ready feature)
10. Powder room RDC: confirm direct supply gives acceptable wait time (~10-15 sec for ~5-7m)

FLOOR PLANS


Entrance & Hallway
3.50 m2
Powder Room
1.50 m2
Laundry
1.80 m2
Living
29.50 m2
Kitchen & Meals
24.70 m2


Bedroom
7.50 m2
Hall
4.05 m2
Bathroom
5.80 m2
Circulation
3.80 m2
Walk-in Closet
9.30 m2
Bedroom
16.70 m2
Bedroom
15.20 m2

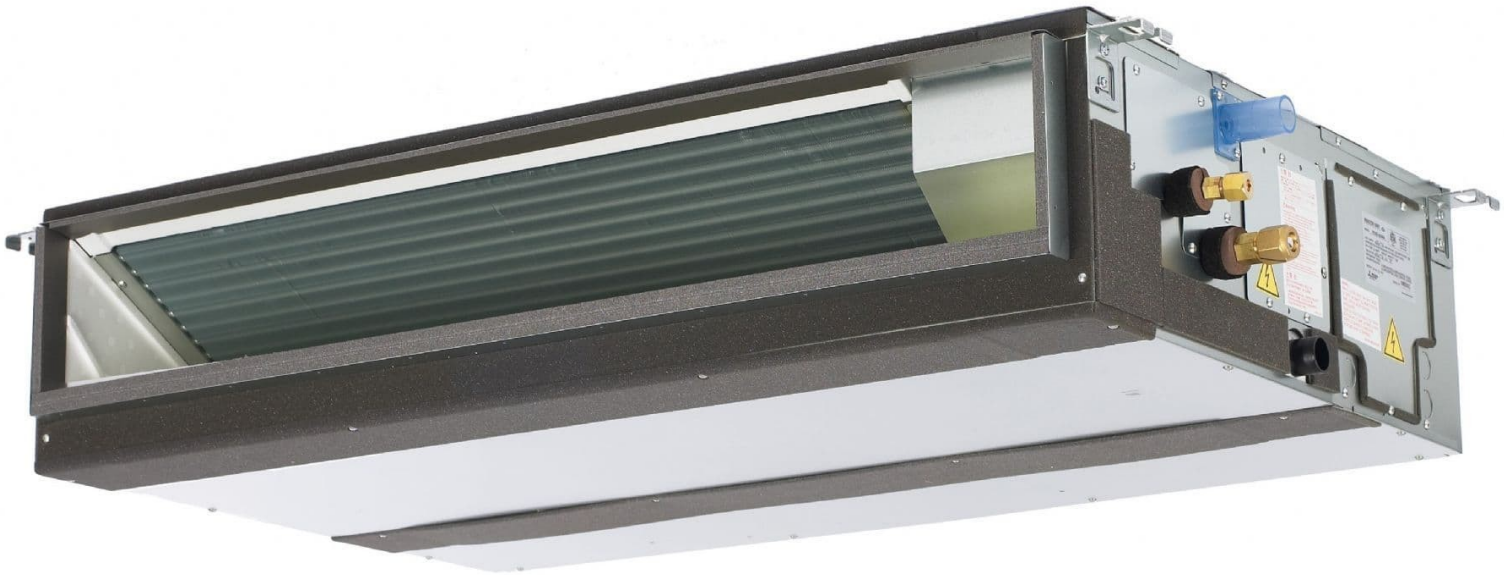


Pre piping before fondation



Équipement Photos

1 PEAD-M71JA (RDC)



Équipement Photos



2 PEAD-M50JA (étage)



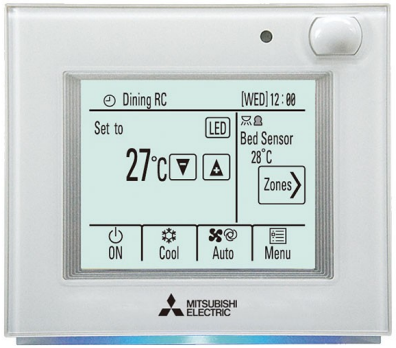
Équipement Photos

3 SUZ-M71VA (unité extérieure)



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Équipement Photos

4	PAC-ZC40H-E (contrôleur de zone)	
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Équipement Photos

5 Lossnay VL-250 (VMC)





Eau — Ariston / Grundfos / Atlas / Bosch :

#	Équipement	Photos
6	Ariston Nuos Plus S2 250L (PAC ECS)	

# Équipement	Photos
7 Grundfos Comfort PM AutoAdapt (pompe)	

# Équipement	Photos
8 Atlas Filtri (filtre sédiments)	 

#	Équipement	Photos
9	bosch-homecomfort.com Descalcificador de agua Aqua 4000 S | Junkers Bosch Aqua 4000 S (adoucisseur Phase 2)	